# NCTU-EE IC Design LAB – 2025Fall

## Final Project: Innovus - From RTL to GDSII

In this exercise, you are going to perform the APR steps to the design: TMIP

## 1. Memory lef file modification

1. Modify all the Memory.lef file

   **a.** Change “core” to “core_5040”

   **b.** Change “ME1” to “metal1” and so on

   **c.** Change “VI1” to “via”

   **d.** Change “VI2” to “via2” and so on

```
NAMESCASESENSITIVE ON ;
MACRO MEM_IMAGE
CLASS BLOCK ;
FOREIGN MEM_IMAGE 0.000 0.000 ;
ORIGIN 0.000 0.000 ;
SIZE 994.480 BY 176.400 ;
SYMMETRY x y r90 ;
SITE core ;
PIN VCC
  DIRECTION INOUT ;
  USE POWER ;
  SHAPE ABUTMENT ;
 PORT
  LAYER ME4 ;
  RECT 993.360 164.980 994.480 168.220 ;
  LAYER ME3 ;
  RECT 993.360 164.980 994.480 168.220 ;
  LAYER ME2 ;
  RECT 993.360 164.980 994.480 168.220 ;
  LAYER ME1 ;
  RECT 993.360 164.980 994.480 168.220 ;
 END
```

```
  LAYER VI1 ;
  RECT 0.000 0.140 994.480 176.400 ;
  LAYER VI2 ;
  RECT 0.000 0.140 994.480 176.400 ;
  LAYER VI3 ;
  RECT 0.000 0.140 994.480 176.400 ;
```

## 2. Data Preparation

1. Change the directory to **05_APR**

2. Prepare chip design netlist

   **a.** Open **CHIP_SHELL.v** and complete the definition of pad cells
   - ➢ The top module name is **CHIP**
   - ➢ This **CHIP** contains the module **TMIP;** and **I/O**, **I/O power**, **core power** pad
   - ➢ Please calculate how many I/O power pads and core power pads you need, and complete the netlist of pad cells

   **b.** After defining the pad cells, please run “**./00_combine**” to combine the MVDM_SYN.v with CHIP_SHELL.v to be CHIP_SYN.v

3. Prepare I/O pad location file:

   **a.** Please see **CHIP.io** to know how to assign I/O pad location.

   **b.** Open **CHIP.io** and complete the I/O pad location assignment according to the netlist of pad cells in **CHIP_SYN.v**

4. Prepare timing/driving/loading constraint file **CHIP.sdc** and **CHIP_cts.sdc**

   **a.** CHIP.sdc:
   - ➢ cp ../02_SYN/Netlist/MVDM_SYN.sdc CHIP.sdc
   - ➢ Comment out the set_wire_load_mode & set_wire_load_model instruction

```
#set_wire_load_mode top
#set_wire_load_model -name G5K -library fsa0m_a_generic_core_ss1p62v125c
```

**b.** CHIP_cts.sdc:

- User should set their own cycle time and delay.

```
####################################################################

# Created by write_sdc on Thu Mar 13 15:04:04 2025

####################################################################
set sdc_version 2.1

set_units -time ns -resistance kOhm -capacitance pF -voltage V -current mA
set_operating_conditions -max WCCOM -max_library                   \
fsa0m_a_generic_core_ss1p62v125c\
                         -min BCCOM -min_library                   \
fsa0m_a_generic_core_ff1p98vm40c

create_clock [get_ports clk]  -period 20  -waveform {0 10}
set_drive 0.1 [all_inputs]
set_load -pin_load 20 [all_outputs]
set_input_delay 5 -clock clk [remove_from_collection [all_inputs] [get_ports clk]]
set_output_delay 5 -clock clk [all_outputs]
```

5. Prepare linked library files:

**a. Timing libraries (LIB)**

- **fsa0m_a_generic_core_ss1p62v125c.lib**
- **fsa0m_a_generic_core_ff1p98vm40c.lib**
- **fsa0m_a_t33_generic_io_ss1p62v125c.lib**
- **fsa0m_a_t33_generic_io_ff1p98vm40c.lib**
- **Memory_WC.lib**
- **Memory_BC.lib**

**b. Physical libraries (LEF)**

- **header6_V55_20ka_cic.lef**
- **fsa0m_a_generic_core.lef**
- **FSA0M_A_GENERIC_CORE_ANT_V55.lef**
- **fsa0m_a_t33_generic_io.lef**
- **FSA0M_A_T33_GENERIC_IO_ANT_V55.lef**
- **BONDPAD.lef**
- **Memory.lef**

**c. RC extraction table/files**

- **u18_Faraday.CapTbl**
- **icecaps.tch**

**d. CeltIC libraries**

- **u18_ss.cdb**
- **u18_ff.cdb**

**e. GDSII layout** (Will not stream out in this course)

- **u18.gds2**
- **u18io3v5v_6lm.gds2**
- **Memory.gds2** (not provided in this compiler version)

## 2. Reading Cell Library information and Netlist for APR

1. If you are running on ee servers, type below command on terminal to set OA_HOME.
   **% setenv OA_HOME /RAID2/cad/cadence/INNOVUS/INNOVUS_20.15.000/share/oa**
   ```
   @ee31[~/Lab11/05_APR]$ setenv OA_HOME /RAID2/cad/cadence/INNOVUS/INNOVUS_20.15.000/share/oa
   ```

2. Start Innovus in the directory **05_APR**
   **% innovus** (no &, do NOT use background execution)
   ```
   @ee31[~/Lab11/05_APR]$ innovus
   ```

3. Set uniquify global variable to 1
   **% set init_design_uniquify 1**
   ```
   innovus 1> set init_design_uniquify 1
   ```

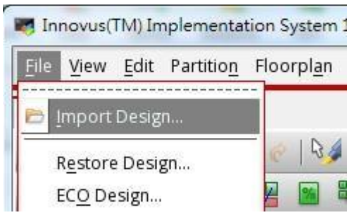
4. Change design mode to 0.18um process
   **% setDesignMode -process 180**
   ```
   innovus 2> setDesignMode -process 180
   ```

5. Skip error messages like the following figure when reading lef files
   **% suppressMessage TECHLIB 1318**
   **% suppressMessage ENCEXT-2799**
   ```
   innovus 3> suppressMessage TECHLIB 1318
   innovus 4> suppressMessage EXCEXT-2799
   ```

```
**ERROR: (TECHLIB-1318):      All the table values in the 'rise_transition' group are within '0
.000010' of each other. (File /home/RAID2/COURSE/iclab/iclabta01/umc018/Lib/umc18io3v5v_slow.lib,
 Line 682)
**ERROR: (TECHLIB-1318):      All the table values in the 'fall_transition' group are within '0
.000010' of each other. (File /home/RAID2/COURSE/iclab/iclabta01/umc018/Lib/umc18io3v5v_slow.lib,
 Line 690)
**ERROR: (TECHLIB-1318):      All the table values in the 'rise_transition' group are within '0
.000010' of each other. (File /home/RAID2/COURSE/iclab/iclabta01/umc018/Lib/umc18io3v5v_slow.lib,
 Line 782)
```

This error message occurs since innovus assumes the values of rise / fall transition time should differ larger than 0.00001 ns in **the table** depending on the supply Voltage or the Temperature **in .lef file**. The error message can be ignored if the given .lef file is ensured to be correct. (Note: only suppress error when the reason is clearly verified instead of suppressing all errors)

6. In innovus menu, open ***File → Import Design***




7. Fill the following field:

**a.** Netlist

◆ Verilog

➢ Files: **CHIP_SYN.v**

➢ Top Cell: ◆ By User : **CHIP**

**b.** Technology / Physical Libraries

◆ LEF Files

➢ header6_V55_20ka_cic.lef

➢ fsa0m_a_generic_core.lef

➢ FSA0M_A_GENERIC_CORE_ANT_V55.lef

➢ **fsa0m_a_t33_generic_io.lef**

➢ **FSA0M_A_T33_GENERIC_IO_ANT_V55.lef**

➢ **BONDPAD.lef**

➢ **Memory.lef**

Note that (1) the standard cell lef file should be put in the first place since in contains the major metals / vias / polys technology definitions. Also (2) if there are antenna lef files, they should be put after the corresponded lef files without describing antenna layers. For example: umc18_6lm.lef should be put in the first place; umc18_6lm_antenna.lef should be put after umc18_6lm.lef.

**c.** Floorplan

➢ IO Assigment Files: **CHIP.io**

**d.** Power

➢ Power Nets: **VCC**

➢ Ground Nets: **GND**

**e.** Analysis Configuration

Press “Create Analysis Configuration” tab

➢ Double click **Library Sets** and include the max and min delay library :

**Max delay:**

Name: lib_max

Timing Library:

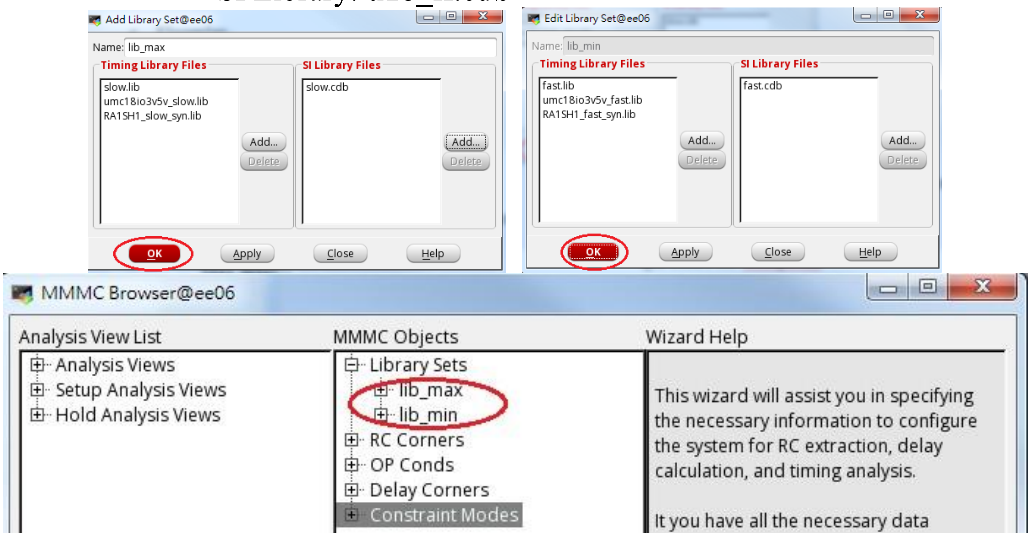
**fsa0m_a_generic_core_ss1p62v125c.lib**

**fsa0m_a_t33_generic_io_ss1p62v125c.lib**

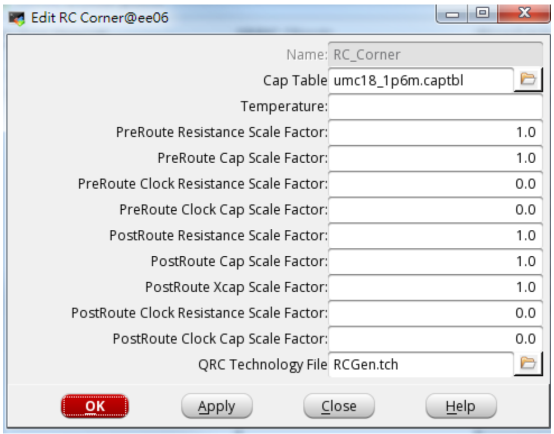
**Memory_WC.lib**

SI Library: **u18_ss.cdb**

**Min delay:**
Name: lib_min
Timing Library:
**fsa0m_a_generic_core_ff1p98vm40c.lib**
**fsa0m_a_t33_generic_io_ff1p98vm40c.lib**
**Memory_BC.lib**
SI Library: **u18_ff.cdb**




➢ Double click **RC Corners** to include the RC corner library :
Name: RC_Corner
Cap Table: **u18_Faraday.CapTbl**
QRC Tech File: **icecaps.tch**

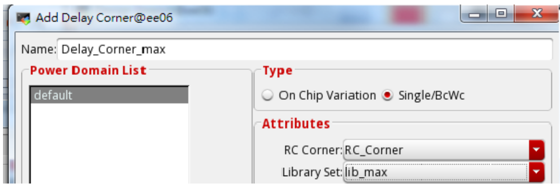


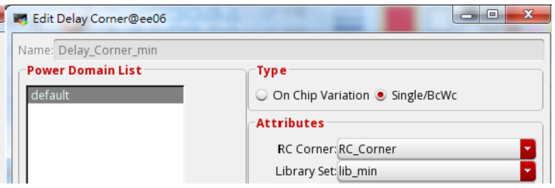

➢ Double click **Delay Corners** and create max and min delay constraints :

**Max delay:**
Name: Delay_Corner_max
RC Corner: RC_Corner
Lib Set: lib_max

**Min delay:**
Name: Delay_Corner_min
RC Corner: RC_Corner

Lib Set: lib_min



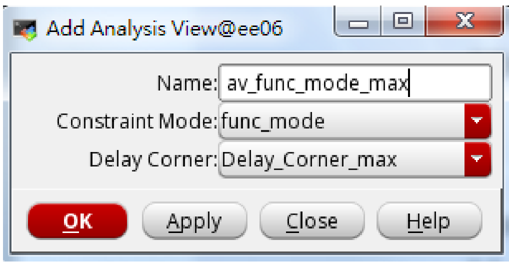



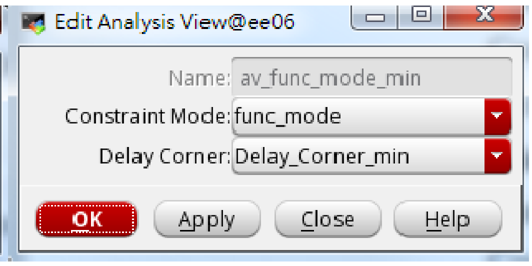

- Double click **Constraints Mode** and create a function mode to place CHIP.sdc :
  Name: func_mode
  SDC Constraint Files: CHIP.sdc


- Double click **Analysis Views** to create max and min delay analysis

  **Max delay:**
  Name: av_func_mode_max
  Constraint Mode: func_mode
  Delay Corner: Delay_Corner_max

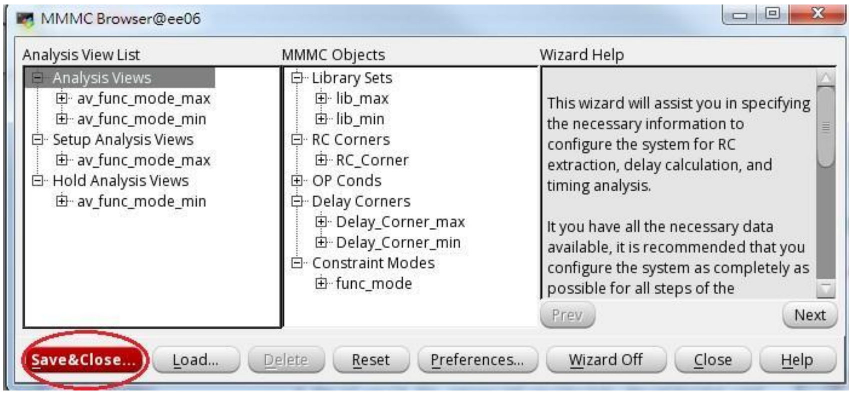
  **Min delay:**
  Name: av_func_mode_min
  Constraint Mode: func_mode
  Delay Corner: Delay_Corner_min



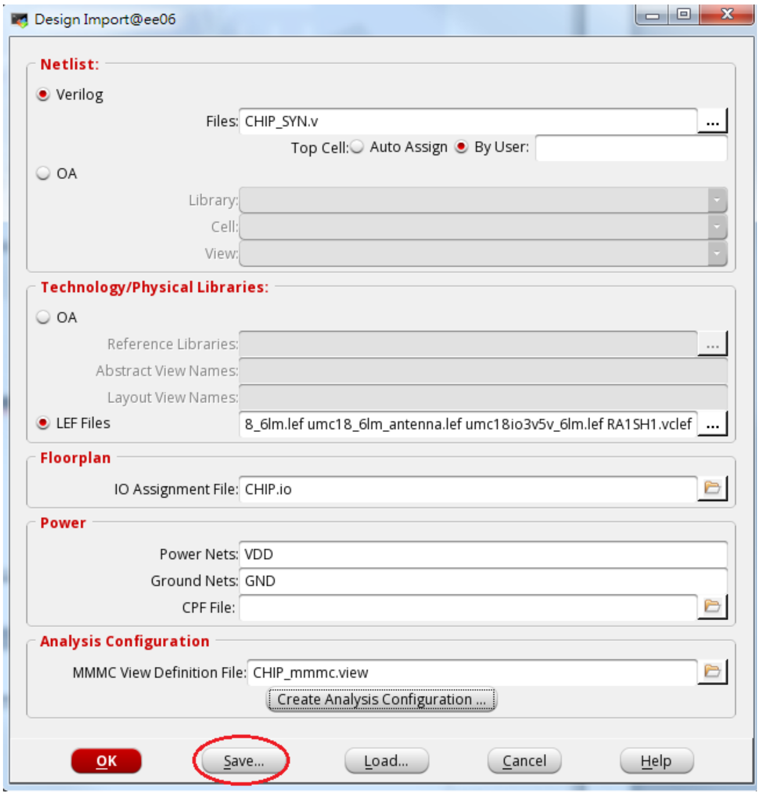




- Click **Setup Analysis View** and specify the max analysis mode
  Choose: av_func_mode_max

- Click **Hold Analysis View** and specify the min analysis mode
  Choose: av_func_mode_min


MMMC Browser@ee06
Analysis View List
MMMC Objects
Wizard Help
Analysis Views
av_func_mode_max
av_func_mode_min
Setup Analysis Views
av_func_mode_max
Hold Analysis Views
av_func_mode_min
Library Sets
lib_max
lib_min
RC Corners
RC_Corner
OP Conds
Delay Corners
Delay_Corner_max
Delay_Corner_min
Constraint Modes
func_mode
This wizard will assist you in specifying the necessary information to configure the system for RC extraction, delay calculation, and timing analysis.
It you have all the necessary data available, it is recommended that you configure the system as completely as possible for all steps of the
Prev
Next
Save&Close...
Load...
Delete
Reset
Preferences...
Wizard Off
Close
Help

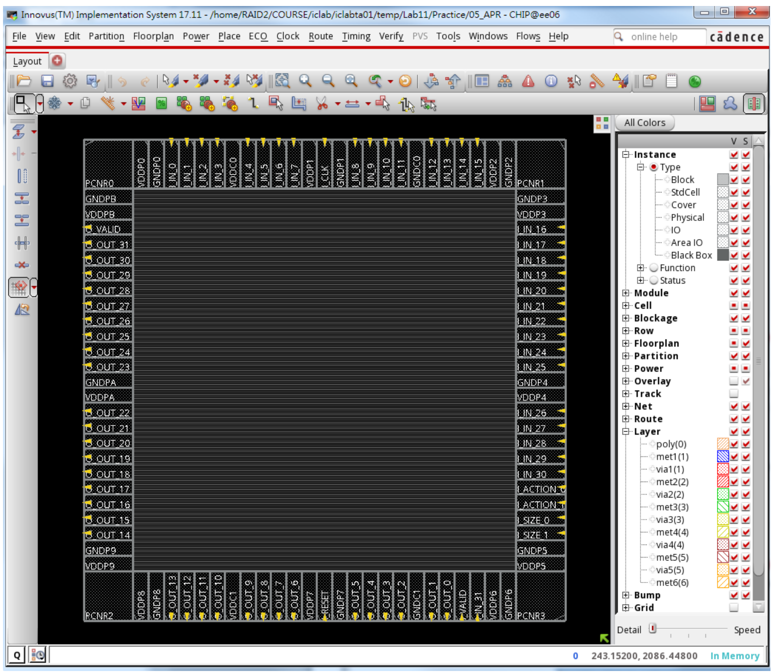
➢ Save as "CHIP_mmmc.view"




8. Save current settings:
   **a.** Click **Save…** button
      ➢ File name: **CHIP.globals**

   If you want to reload the settings, you can just press Load… button

9. Click **OK** button



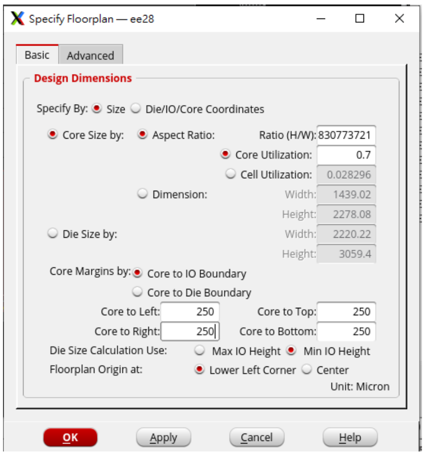
## 3. Specify Chip Floorplan

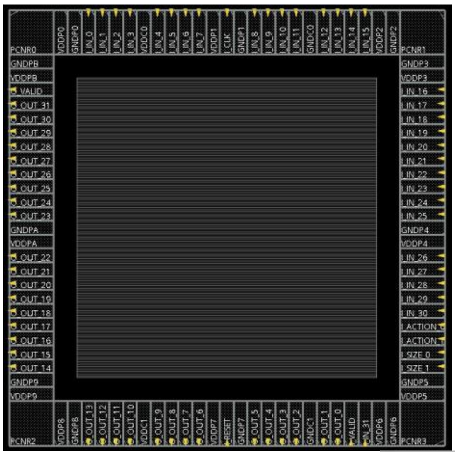
1. In innovus menu, open ***Floorplan → Specify Floorplan***

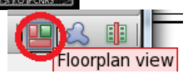
2. Specify core size:
   a. Core Utilization: Set any as your wish (0.7~0.8 is suggested)

3. Specify core margin:
   b. ◆ Core to IO Boundary
   Core to Left: 100
   Core to Right: 100
   Core to Top: 100
   Core to Bottom: 100

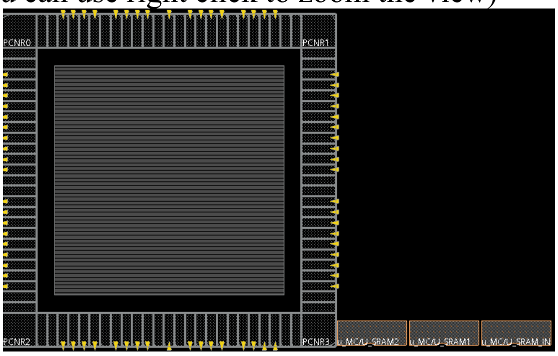
4. Apply the specification:
   c. Click **OK** button



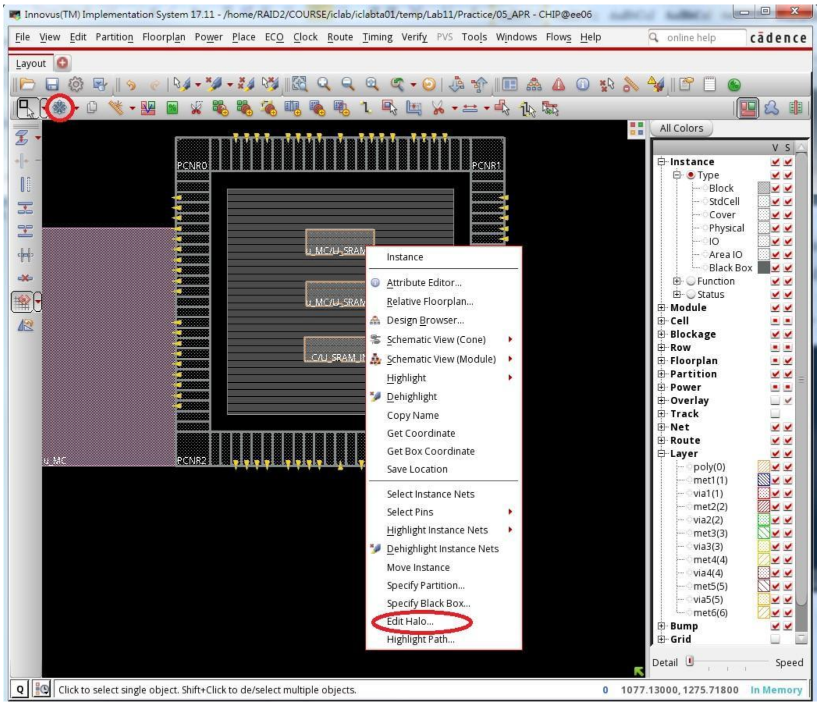


5. You can view designs in three ways; change to floorplan view: 

   The memory macro should appear (Besides zoom in and zoom out button , you can use right click to zoom the view)



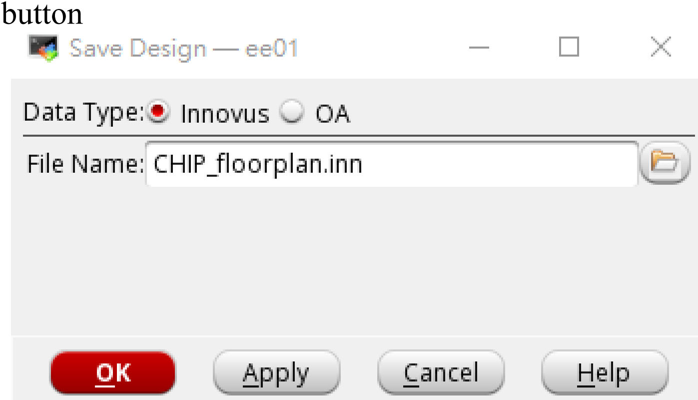
6. Move them to the desired places and make placement blockage: right click on one macro, Choose "**Edit Halo**"




7. Specify Halo For: ◆ All Macros
   Add/Update Halo: Top, Bottom, Left, Right: 15um

8. You can also create the placement blockage yourself with the button:


9. Save the floorplan design:
   **a.** File → Save Design
   **b.** Data Type ◆ Innovus
   File Name: CHIP_floorplan.inn
   **c.** Click **OK** button




If you do the wrong things in the following steps such as power planning, you can restore the design from the previous steps. However if the input files such as verilog file / io file / timing constraints are changed, you have to rerun all the APR steps.

## 4. Connect/Define Global Net

1. In innovus menu, open ***Power → Connect Global Nets…***


2. Add all VDD pins to Connection List:
    **a.** Connect ◆ Pin

    Instance Basename: *
    Pin Name(s): **VCC**
    **b.** Scope ◆ Apply All
    **c.** To Global Nets: VCC

    **d.** Click Add to List button

3. Add all VDD nets to Connection List:
    **a.** Connect ◆ Net Basename: **VCC**

    **b.** Scope ◆ Apply All
    **c.** To Global Nets: VCC

    **d.** Click Add to List button

4. Add all Tie High pins to Connection List:
    **a.** Connect ◆ Tie High

    **b.** Scope ◆ Apply All
    **c.** To Global Nets: VCC

    **d.** Click Add to List button

5. Add all GND pins to Connection List:
    **a.** Connect ◆ Pin

    Instance Basename: *
    Pin Name(s): **GND**
    **b.** Scope ◆ Apply All
    **c.** To Global Nets: GND

    **d.** Click Add to List button

6. Add all GND nets to Connection List:
    **a.** Connect ◆ Net Basename: **GND**

    **b.** Scope ◆ Apply All
    **c.** To Global Nets: GND

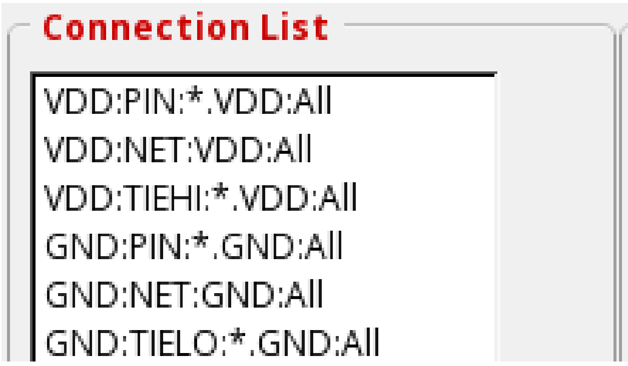
    **d.** Click Add to List button

7. Add all Tie Low pins to Connection List:
    **a.** Connect ◆ Tie Low

    **b.** Scope ◆ Apply All
    **c.** To Global Nets: GND

d. Click Add to List button

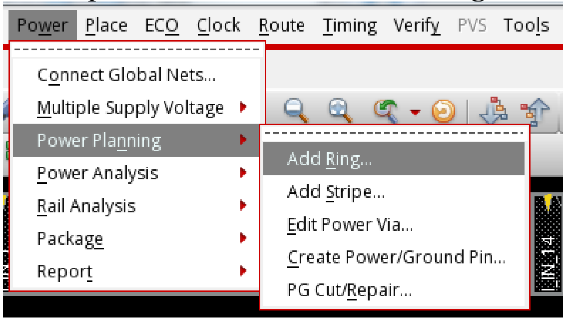
e. Click Add to List button




8. Apply the connection list and check:
   e. Click **Apply** button
   f. Click **Check** button
   g. Click **Cancel** button

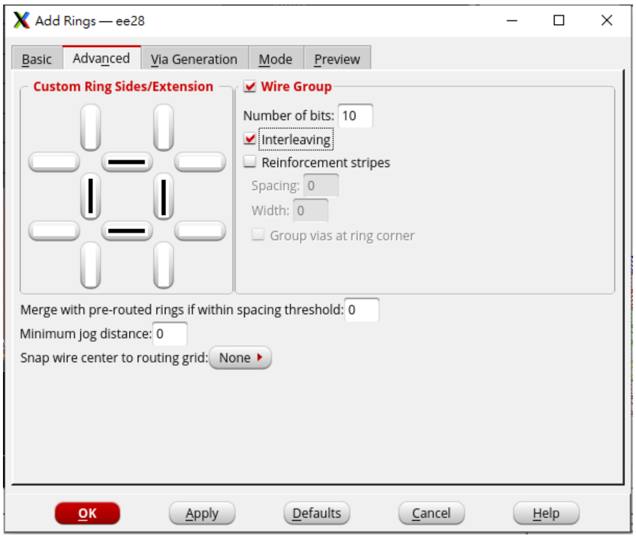
## 5. Power Planning (Add Core Power Rings)

1. In innovus menu, open ***Power → Power Planning → Add Rings…***

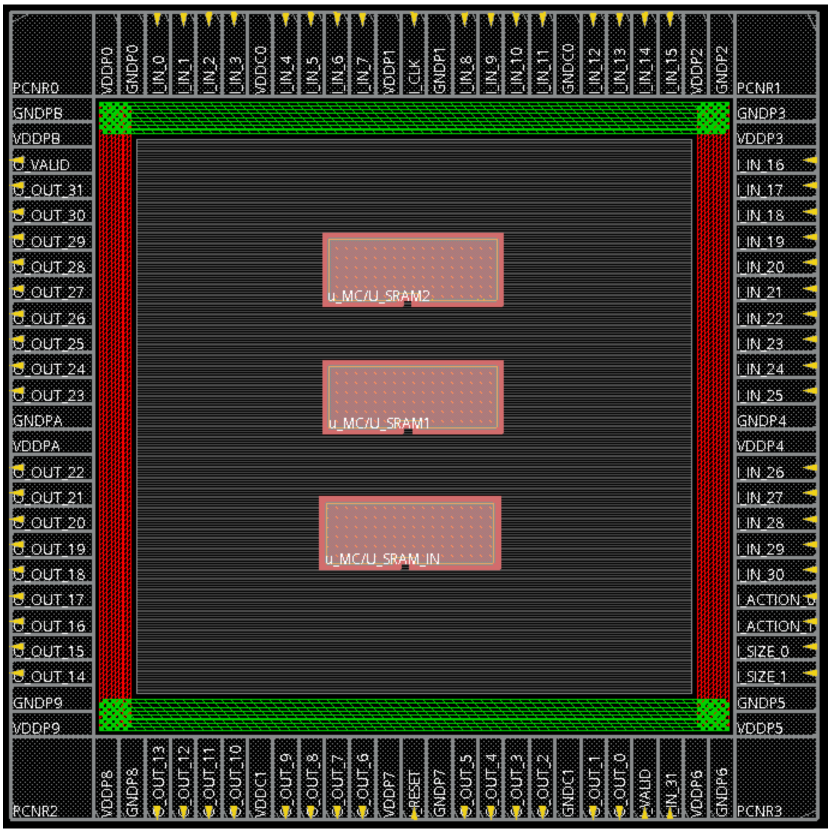



2. In the **Basic** tab, fill the following field:
   a. Net(s): GND VCC
   b. Ring Type: Core ring(s) contouring
   c. Specify metal layers and width
      - Top Layer: **metal3 H** Width: **9**
      - Bottom Layer: **metal3 H** Width: **9**
      - Left Layer: **metal2 V** Width: **9**
      - Right Layer: **metal2 V** Width: **9**
      - ◆ Offset: Center in channel
      - Click **Update** button


3. In the **Advanced** tab, fill the following field:
   **a.** ◆ Wire Group
   **b.** Number of bits: **5**
   **c.** ◆ Interleaving
4. Click **OK** button



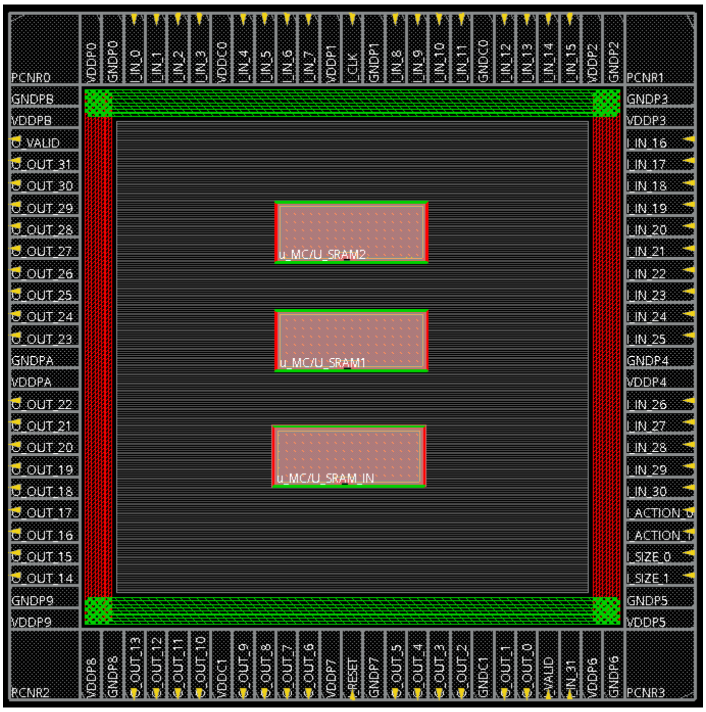

5. Check if the ring is correctly created. If not, click **undo** button  and repeat step 2~4 again.




## 6. Power Planning (Add Block Rings)

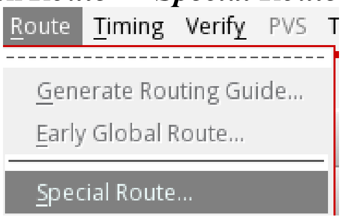
**1.** In innovus menu, open ***Power → Power Planning → Add Rings…***

2. In the **Basic** tab, fill the following field:
   **a.** Net(s): GND VCC
   **b.** Ring Type: Block ring(s) around
   **c.** Specify metal layers and width
      - ➢ Top Layer: **met3(3) H** Width: **2**
      - ➢ Bottom Layer: **met3(3) H** Width: **2**
      - ➢ Left Layer: **met2(2) V** Width: **2**
      - ➢ Right Layer: **met2(2) V** Width: **2**
      - ➢ ◇ Offset: Center in channel
      - ➢ Click **Update** button
3. In the **Advanced** tab, disable the wire group:
   **a.** ◇ Wire Group
4. Click **OK** button




## 7. Connect Core Power Pin

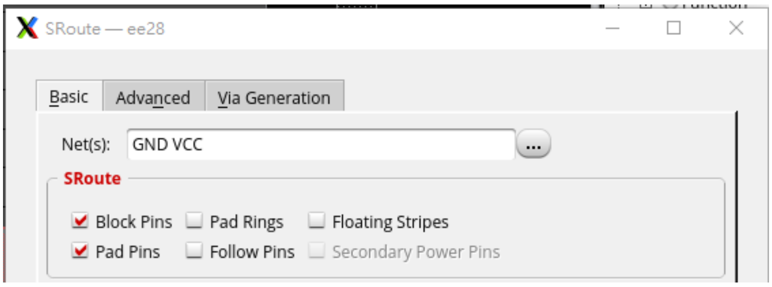
*1.* In innovus menu, open ***Route → Special Route…***

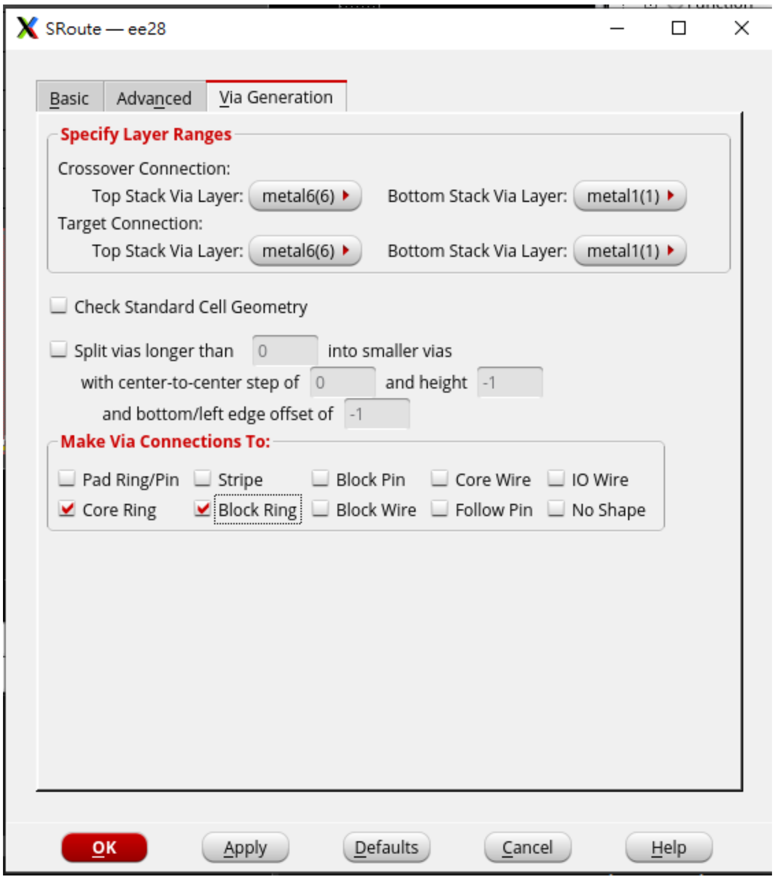


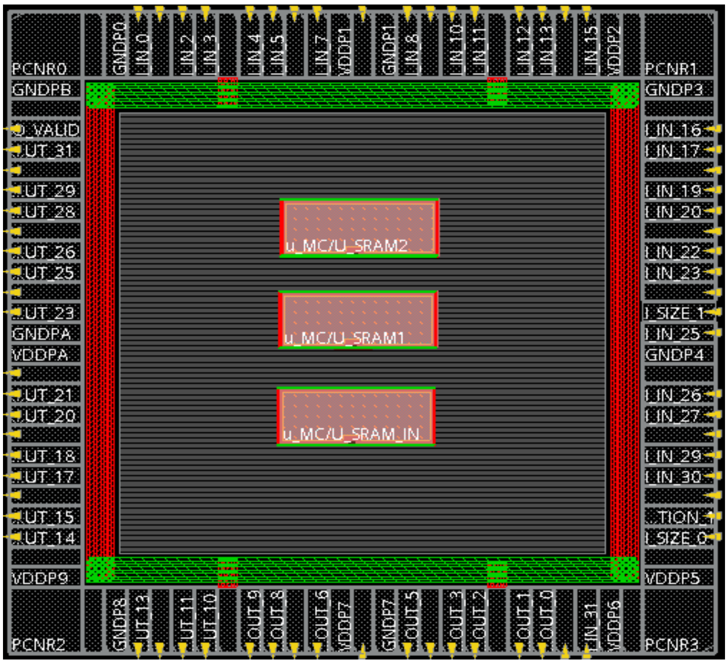

2. Connect core power:
   In **Basic** tab,
   **a.** Net(s): GND VCC
   **b.** Set the following configuration
   **SRoute**:
   - ◆ Block pins
   - ◇ Pad rings
   - ◇ Floating Stripes
   - ◆ Pad pins
   - ◇ Follow pins

   In **Via Generation** tab,
   **c.** Set the following configuration
   - ◆ Core Ring
   - ◆ Block Ring

   **d.** Click **OK** button

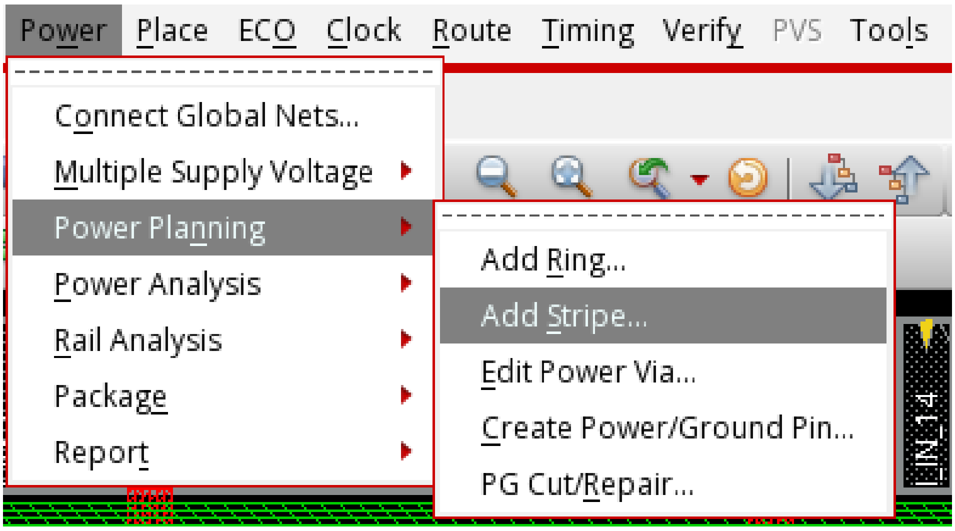

SRoute — ee28
Basic
Advanced
Via Generation
Net(s):
GND VCC
SRoute
Block Pins
Pad Rings
Floating Stripes
Pad Pins
Follow Pins
Secondary Power Pins


SRoute — ee28
Basic
Advanced
Via Generation
Specify Layer Ranges
Crossover Connection:
Top Stack Via Layer: metal6(6)
Bottom Stack Via Layer: metal1(1)
Target Connection:
Top Stack Via Layer: metal6(6)
Bottom Stack Via Layer: metal1(1)
Check Standard Cell Geometry
Split vias longer than 0 into smaller vias
with center-to-center step of 0 and height -1
and bottom/left edge offset of -1
Make Via Connections To:
Pad Ring/Pin
Stripe
Block Pin
Core Wire
IO Wire
Core Ring
Block Ring
Block Wire
Follow Pin
No Shape
OK
Apply
Defaults
Cancel
Help


PCNR0
PCNR1
PCNR2
PCNR3
GNDPB
GNDP3
GNDPA
VDDPA
GNDP4
VDDP9
VDDP5
GNDP0
VDDP1
GNDP1
VDDP2
GNDP8
VDDP7
GNDP7
VDDP6
u_MC/U_SRAM2
u_MC/U_SRAM1
u_MC/U_SRAM_IN

## 8. Power Planning (Add Stripes)

*1.* In innovus menu, open ***Power → Power Planning → Add Stripes…***

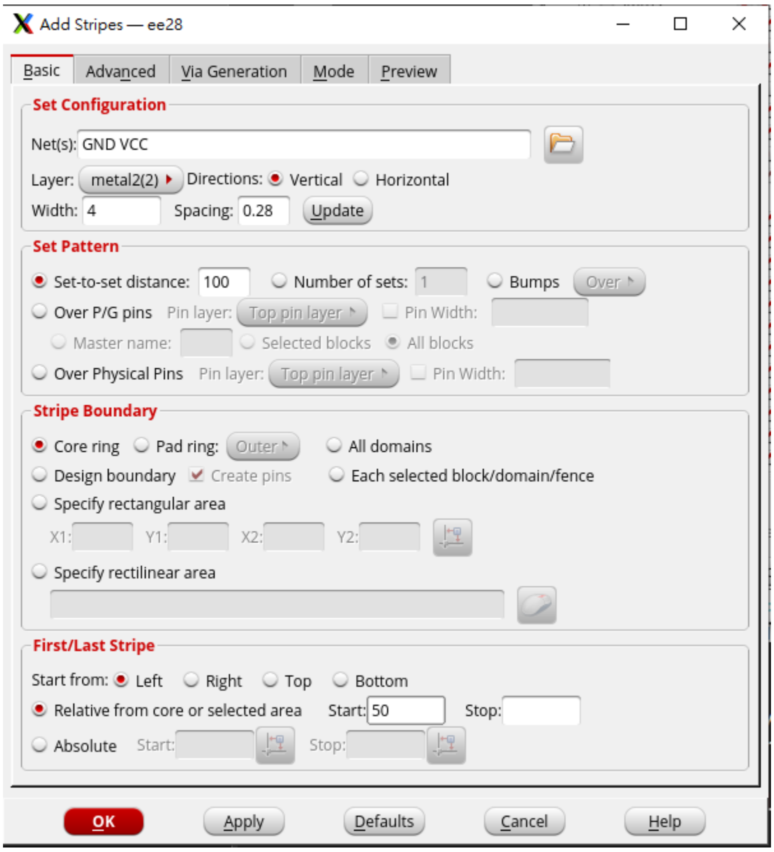


2. Create vertical power stripes:
   **a.** Set Configuration
      - ➢ Net(s): GND VCC
      - ➢ Layer: **met2(2)**
      - ➢ Directions: ◆ Vertical
      - ➢ Width: **4**
      - ➢ Click **Update** Button

   **b.** Set Pattern
      - ➢ ◆ Set-to-set distance: **100**

   **c.** First/Last Stripe
      - ➢ Start from: ◆ Left
      - ➢ ◆ Relative from core or selected area
      - ➢ Start: 60

   **d.** Click **OK** button




3. Check if the stripes are correctly created. If not, click **undo** button  and repeat step a~d again.
4. Create horizontal stripes:
   **a.** Set Configuration
      - ➢ Net(s): GND VCC
      - ➢ Layer: **met3(3)**
      - ➢ Directions: ◆ Horizontal
      - ➢ Width: **4**

- ➢ Click **Update** Button

**b.** Set Pattern

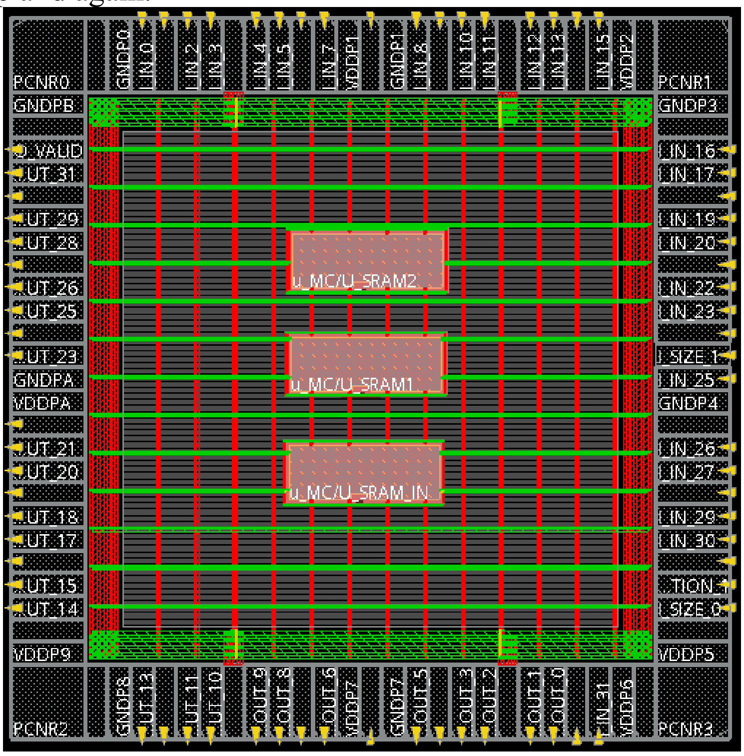
- ➢ ◆ Set-to-set distance: **100**

**c.** First/Last Stripe

- ➢ Start from: ◆ Bottom
- ➢ ◆ Relative from core or selected area
- ➢ Start: 60

**d.** Click **OK** button

5. Check if the stripes are correctly created. If not, click **undo** button and repeat step a~d again.




## 9. Connect Standard Cell Power Line

1. Make sure hard macro blockage has been added in floorplan.
2. In innovus menu, open ***Route → Special Route…***
3. Connect core power:

In **Basic** tab,

**a.** Net(s): GND VCC

**b.** Set the following configuration

- ➢ ◇ Block pins
- ➢ ◇ Pad rings
- ➢ ◇ Floating Stripes
- ➢ ◇ Pad pins
- ➢ ◆ Follow pins

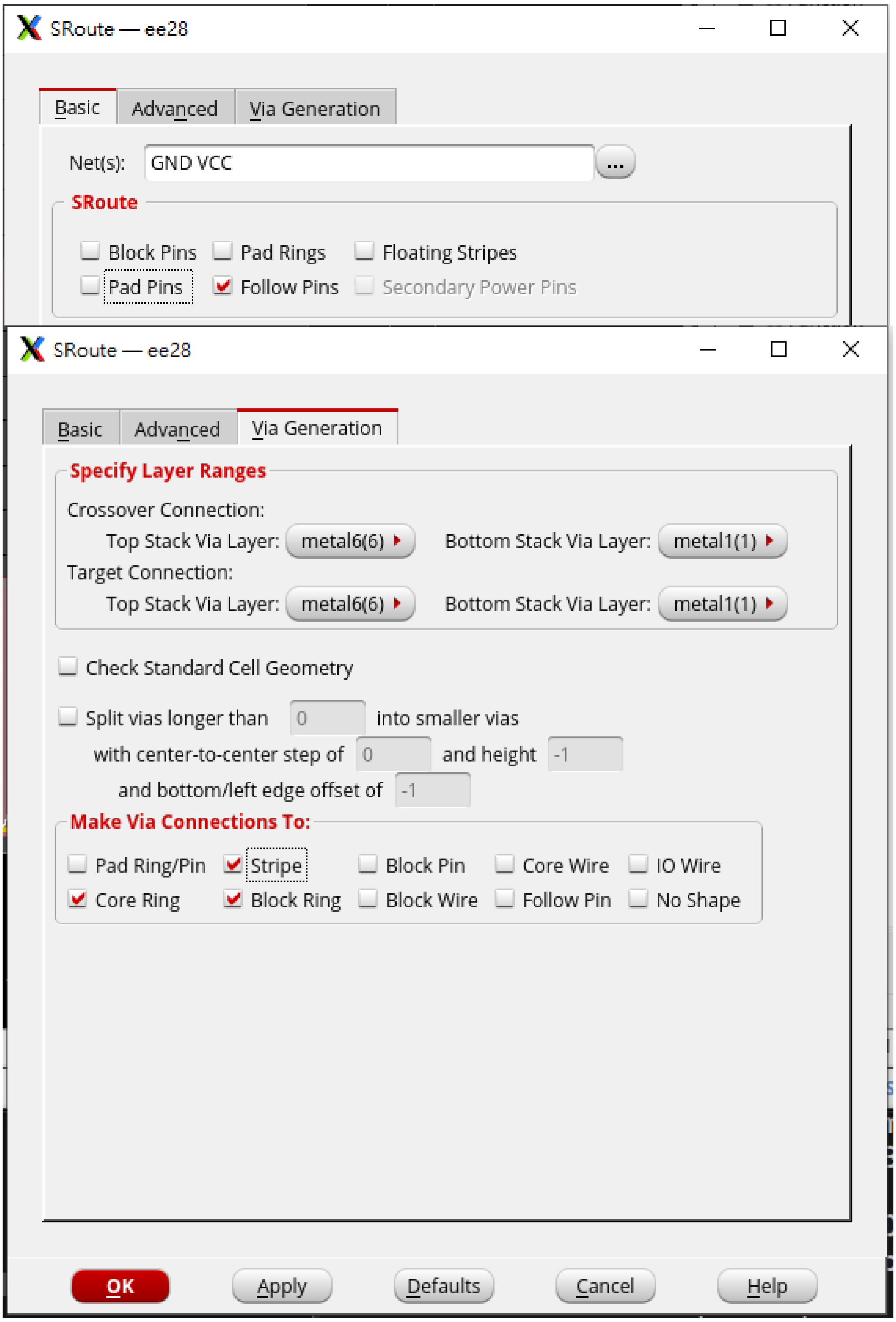
In **Via Generation** tab,

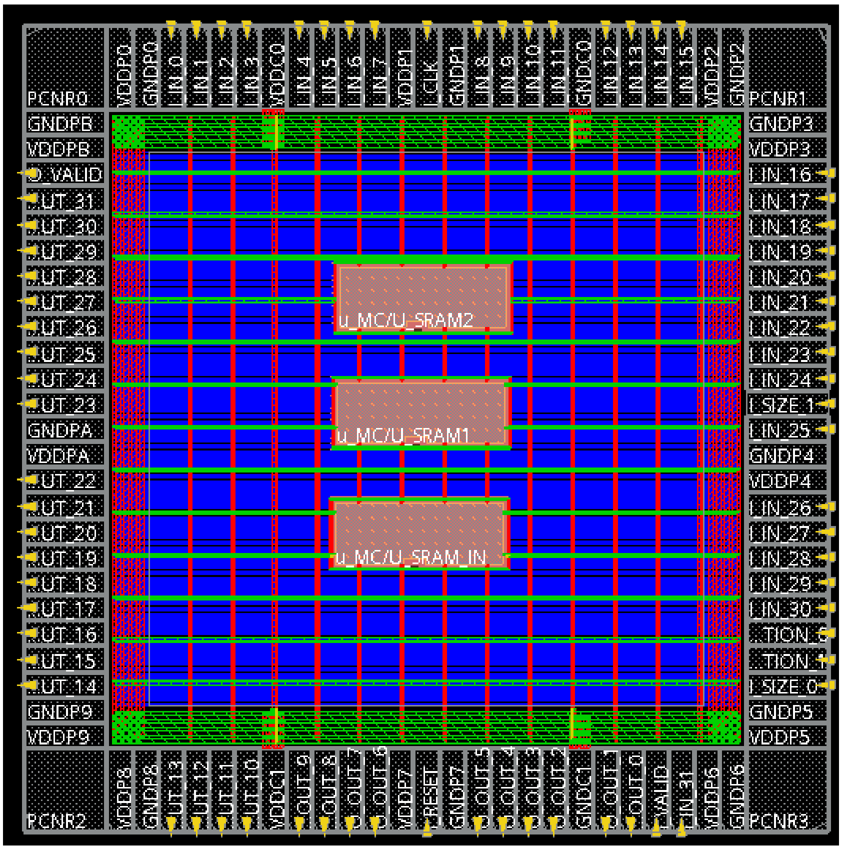
**c.** Set the following configuration

- ➢ ◆ Stripe

- ◆ Core Ring
- ◆ Block Ring

**d.** Click **OK** button






# 10. Verify DRC and LVS

*1.* In innovus menu, open ***Verify → DRC***

- ➢ Click **OK** button
- ➢ Check routing for DRC error

```
Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:00.3  ELAPSED TIME: 0.00  MEM: 1.0M) ***
```

Use ***Tools → Violation Browser*** to help finding Violations locations.

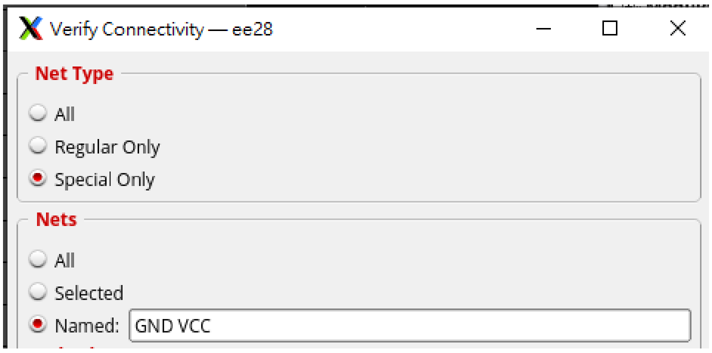
Ex.

Cut_Short via4 → delete the via4 which make this violation

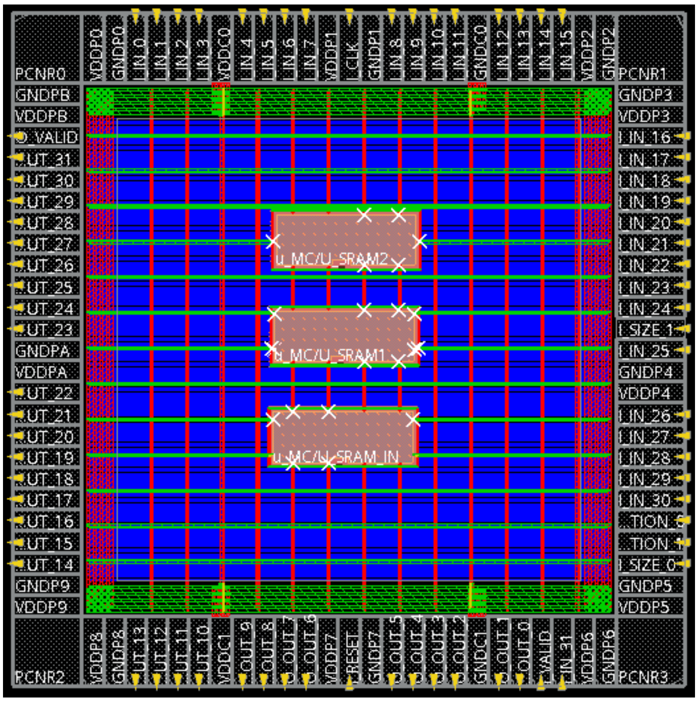
Cut_Spacing via4 → delete the via4 which make this violation

*2.* In innovus menu, open ***Verify → Connectivity***

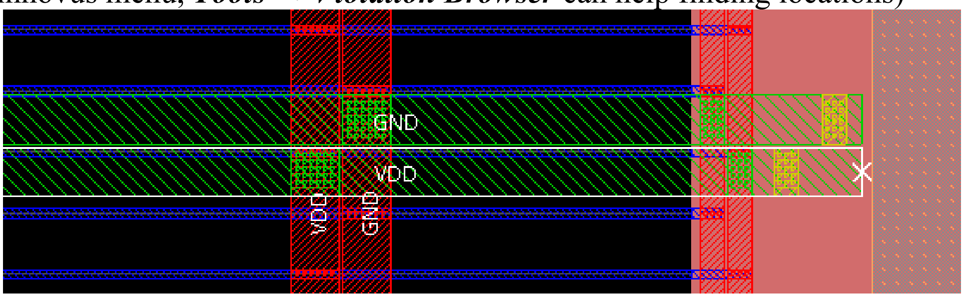
- ➢ Net Type: ◆ Special Only
- ➢ Nets: ◆ Named: GND VCC
- ➢ Click **OK** button
- ➢ Check routing for LVS error



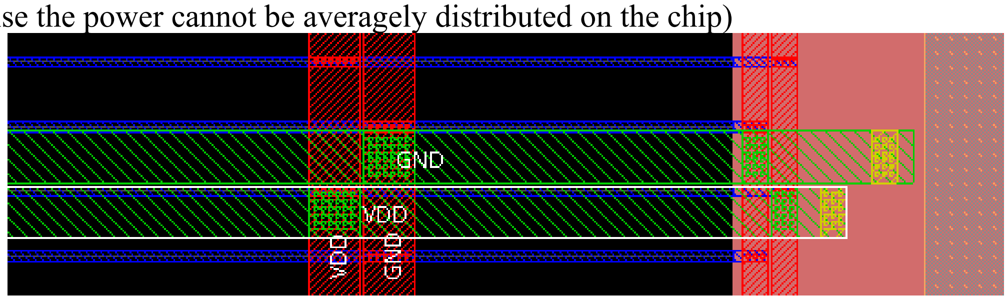

3. If the problem is the dangling wire (floating wire) around memory, like:




Zoom in and left click to select the highlighted wire:
(In innovus menu, ***Tools*** **→** ***Violation Browser*** can help finding locations)




Press "shift+t" , the dangling wire will be adjusted. (Do not delete those wires or else the power cannot be averagely distributed on the chip)




4. After fixing all the dangling wires, run step 2. Again to ensure the

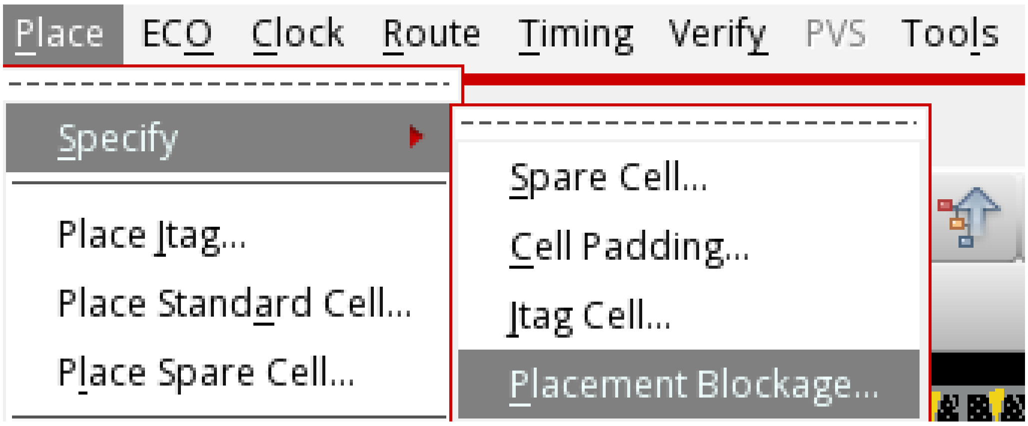
connectivity correctness.

```
******** End: VERIFY CONNECTIVITY ********
 Verification Complete : 0 Viols.  0 Wrngs.
 (CPU Time: 0:00:00.0  MEM: 0.000M)
```

5. Save design as CHIP_powerplan.inn

## 11. Place Standard Cells

*1.* In innovus menu, open ***Place → Specify → Placement Blockage***



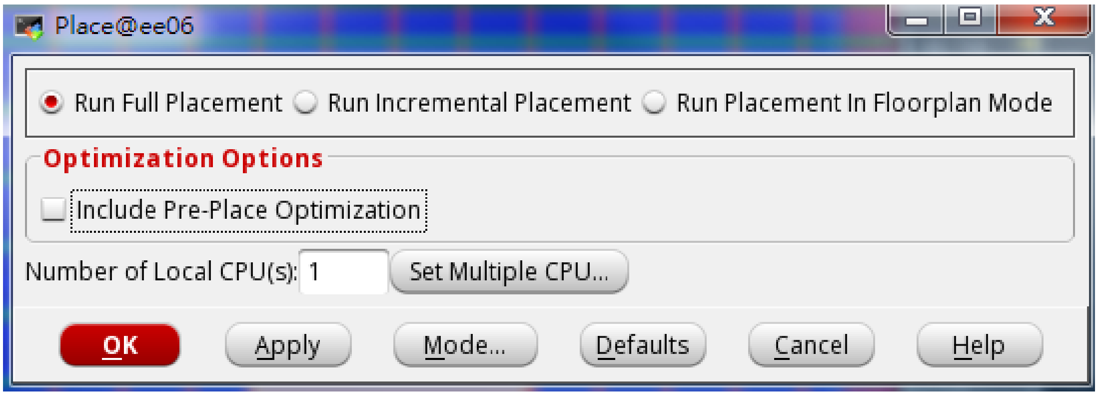
2. Specify placement blockage for stripes
   **a.** Specify placement blockage under metal2 and metal3
      - ◇ M1
      - ◆ M2
      - ◆ M3
      - ◇ M4
      - ◇ M5
      - ◇ M6

   **b.** Click **OK** button

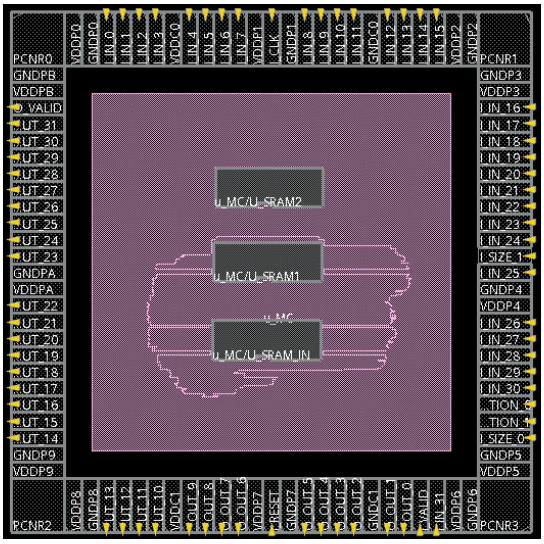

*3.* In innovus menu, open ***Place → Place Standard Cells…***
   - ◆ Run Full Placement

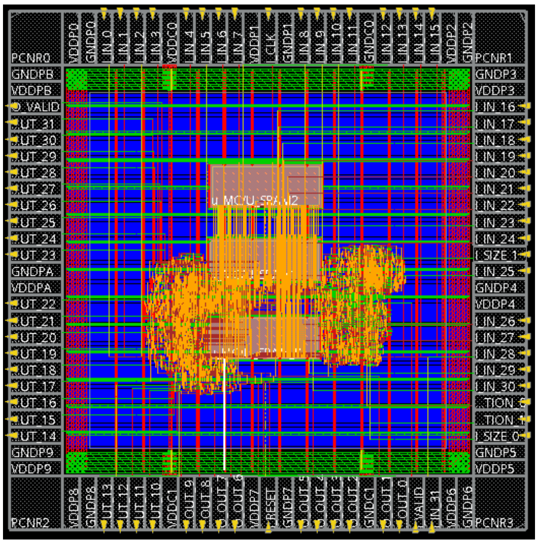
   Optimization Options
   - ◇ Include Pre-Place Optimization

   Click **OK** button




The following figures show the results of Amoeba view and Physical view



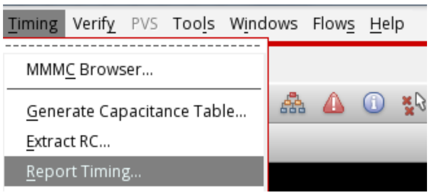


4. Save design as CHIP_placement.inn

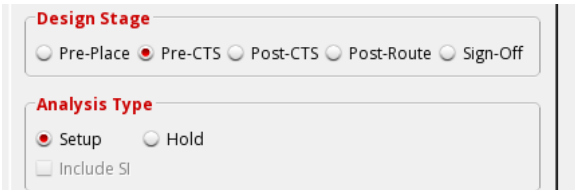
# 12. In-Place Optimization (IPO)

## - Before Clock Tree Synthesis

*1.* In innovus menu, open ***Timing → Report Timing…***



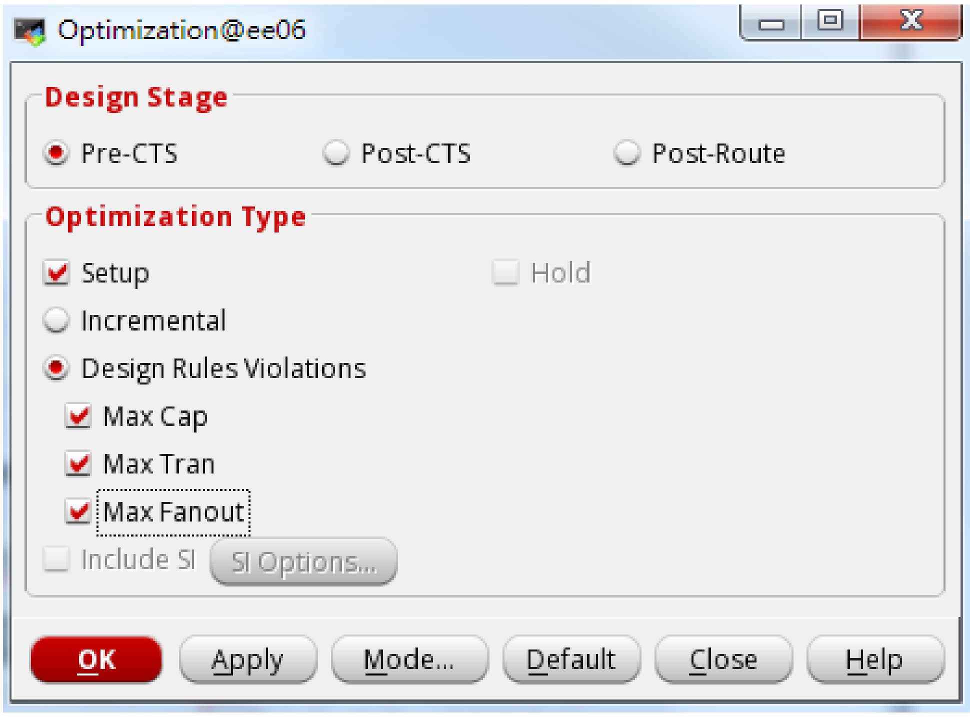

2. Perform trial route to model the interconnection RC effects
   **a.** Design Stage ◆ pre-CTS
   **b.** Analysis Type ◆ Setup
   **c.** Click **OK** button




3. See timing reports in **timingReports/** directory, **CHIP_preCTS.slk** shows the timing analysis results. All slack values must be positive value in this file. Moreover, for detail path report, see **CHIP_preCTS_all.tarpt**. DRVs report files: ***.cap**, ***.fanout**, and ***.tran**.
4. If the timing slack is negative, or there are DRVs, open ***ECO → Optimize Design…*** in innovus menu
5. Perform pre-CTS IPO
   **a.** Design Stage ◆ Pre-CTS
   **b.** Optimization Type
      ➢ ◆ Setup
      ➢ ◆ Design Rule Violations
         ◆ Max Cap
         ◆ Max Tran

◆ Max Fanout

**c.** Click **OK** button



6. Save design as CHIP_preCTS.inn

In preCTS timing report, the max cap violation of DRV may be caused by reset_n or clk, which connects to a lot of registers. After ECO, the rst_n DRV may be fixed by inserting buffers. Whereas the clk may not since the clk buffers will be inserted in the next step: Clock Tree Synthesis (CTS) stage. The remaining clk DRV will be shown like:

| DRVs | Real | | Total |
|---|---|---|---|
| | Nr nets(terms) | Worst Vio | Nr nets(terms) |
| max_cap | 0 (0) | 0.000 | 1 (1) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |

and can be view in the file ***timingReports/CHIP_preCTS.tran.gz***.This DRV should be fixed after synthesizing the clock tree, thus don't worry in this stage. You only have to worry if this DRV cannot be fixed after CTS.

You will also see **Density** and **Routing Overflow** terms at the bottom of the timing report.

```
Density: 8.473%
Routing Overflow: 0.00% H and 0.00% V
```

The lower the density is, the lower the standard cells (excluding Hardmacros) are utilizing the core, which means it provides larger flexibility in further routing and optimization steps.

After the placement and in the preCTS stages, innovus performs **trial route** to give the rough delay calculation of path between every registers / input output ports so the preCTS can be done. Similar to the density which is the placement utilization, the routing overflow represents the wiring congestion level. The higher the routing overflow is, the harder of the coming CTS and detail routing (**nanoroute**). Usually when Horizontal and Vertical routing overflow are both < 0.5% ~ 1.0%, the further routing problem will be small. The routing overflow term will disappear after **nanorouting** since the **detail route** will be given to replace the **trial route**.

## 13. Clock Tree Synthesis (CTS)

In innovus command prompt, execute the following commands:

➢ source ./cmd/ccopt.cmd

## 14. In-Place Optimization (IPO)

### - After Clock Tree Synthesis

***1.*** In innovus menu, open ***Timing → Report Timing…***

2. Perform trial route to model the interconnection RC effects
   **a.** Design Stage ◆ Post-CTS
   **b.** Analysis Type ◆ Setup
   **c.** Click **OK** button
3. See timing reports in **timingReports/** directory, **CHIP_postCTS.slk** shows the timing analysis results. All slack values must be positive value in this file. Moreover, for detail path report, see **CHIP_postCTS_all.tarpt**. DRVs report files: ***.cap**, ***.fanout**, and ***.tran**.
4. If the timing slack is negative, or there are DRVs, open ***ECO → Optimize***

***Design…*** in innovus menu

5. Perform post-CTS IPO
   **a.** Design Stage ◆ Post-CTS
   **b.** Optimization Type
      ➢ ◆ Setup
      ➢ ◆ Design Rule Violations
         ◆ Max Cap
         ◆ Max Tran
         ◆ Max Fanout
   **c.** Click **OK** button

**From post CTS steps, hold time checking is also required**

*6.* In innovus menu, open ***Timing → Report Timing…***
7. Perform trial route to model the interconnection RC effects
   **a.** Design Stage ◆ Post-CTS
   **b.** Analysis Type ◆ Hold
   **c.** Click **OK** button
8. See timing reports in **timingReports/** directory, **CHIP_postCTS_hold.slk** shows the timing analysis results. All slack values must be positive value in this file. Moreover, for detail path report, see **CHIP_postCTS_all_hold.tarpt**. DRVs report files: ***.cap**, ***.fanout**, and ***.tran**.
9. If the timing slack is negative, or there are DRVs, open ***ECO → Optimize Design…*** in innovus menu
10. Perform Post-CTS IPO
   **a.** Design Stage ◆ Post-CTS
   **b.** Optimization Type
      ➢ ◆ Hold
      ➢ ◆ Design Rule Violations
         ◆ Max Cap
         ◆ Max Tran

◆ Max Fanout

c. Click **OK** button

11. Save design as CHIP_postCTS.inn

# 15. Add PAD Filler

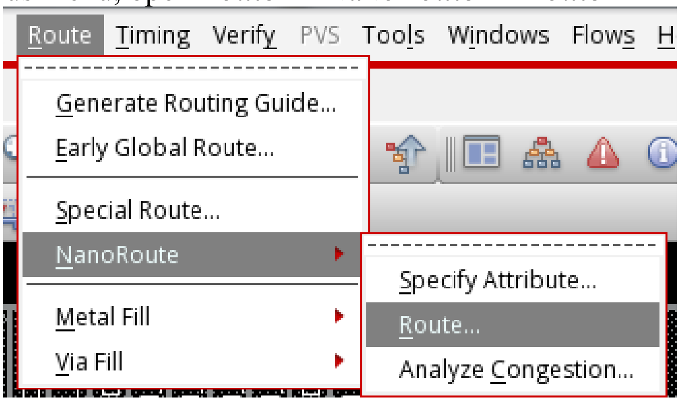
1. In innovus command prompt, execute the following commands:

➢ **source ./cmd/addIOFiller.cmd**

**-fillAnyGap** PAD filler must be added before detail route, or there may have some DRC/LVS violations after PAD filler insertion

# 16. SI-Prevention Detail Route (NanoRoute)

***1.*** In innovus menu, open ***Route → NanoRoute → Route***



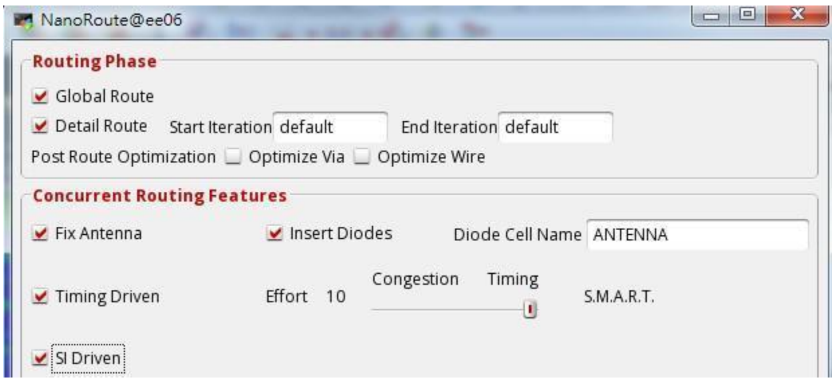

2. Nanoroute can prevent cross talk effects and fix antenna rule violations, also it routes design to meet timing constraints.
   a. Configure routing features

➢ ◆ Fix Antenna

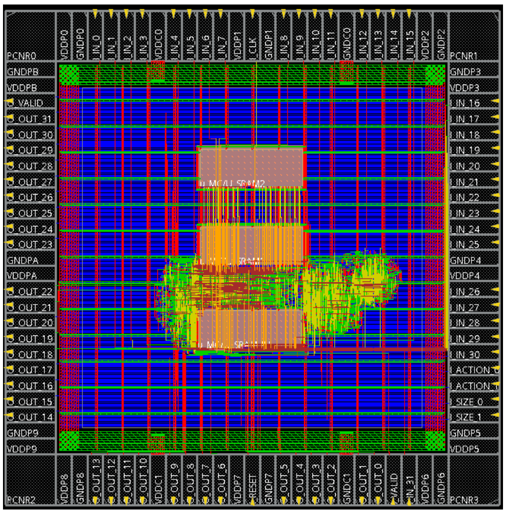
➢ ◆ Insert Diodes Diode Cell Name: ANTENNA

➢ ◆ Timing Driven Effort: 10

➢ ◆ SI Driven

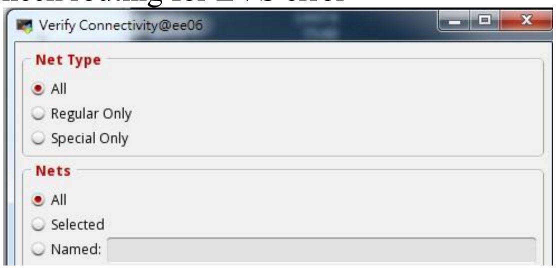



b. Click **OK** button



*3.* In innovus menu, open ***Verify*** → ***Connectivity***

- ➢ Net Type: ◆ All
- ➢ Nets: ◆ All
- ➢ Click **OK** button
- ➢ Check routing for LVS error




*4.* In innovus menu, open ***Verify*** → ***DRC***

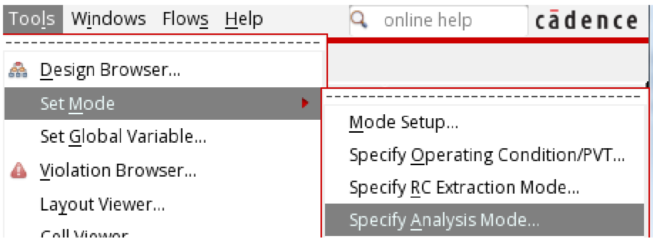
- ➢ Click **OK** button
- ➢ Check routing for DRC error


5. Save design as CHIP_nanoRoute.inn

# 17. In-Place Optimization (consider crosstalk effects)

## - After Detail Route

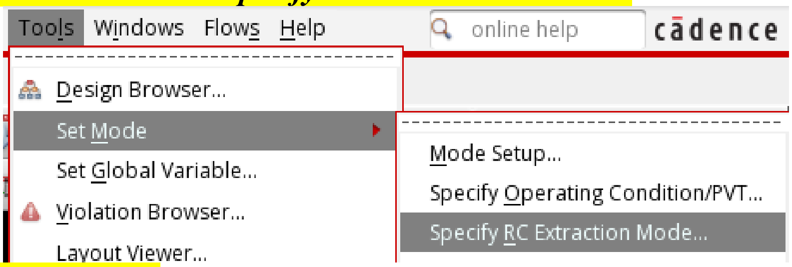
*1.* In innovus menu, open ***Tools*** → ***Set Mode*** → ***Specify Analysis Mode…***




**a.** Timing Mode: ◆ On-Chip Variation

**b.** Click **OK** button


2. In innovus menu, open **_Tools → Set Mode → Specify RC Extraction Mode…_**




   **a.** ◆ PostRoute

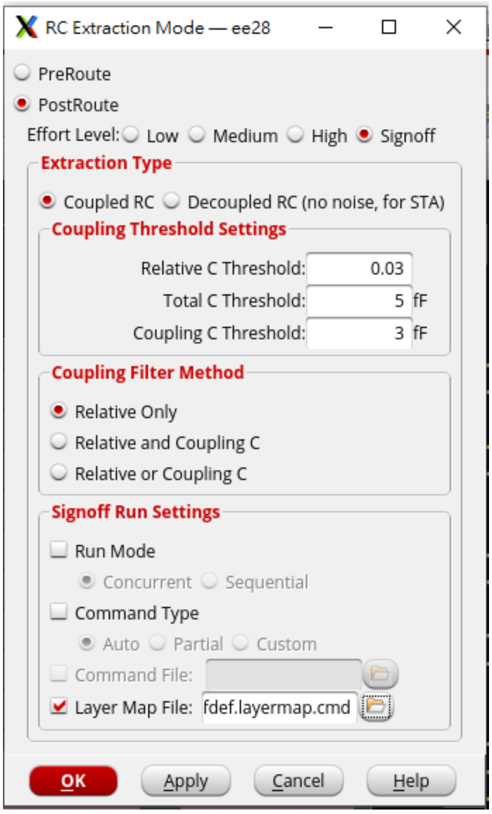
   **b.** Effort Level: ◆ Signoff

   **c.** Extraction Type: ◆ Coupled RC

   **d.** Signoff Run Settings:

◆ Layer Map File: lefdef.layermap.cmd

   **e.** Click **OK** button




***3.*** Setting rc and si

- ➢ innovus > set_db extract_rc_engine post_route
- ➢ innovus > set_db extract_rc_effort_level high
- ➢ innovus > set_db delaycal_enable_si true

***4.*** In innovus menu, open ***Timing → Report Timing…***

5. Perform trial route to model the interconnection RC effects

   **a.** Design Stage ◆ Post-Route

   **b.** Analysis Type ◆ Setup

c. Click **OK** button

6. See timing reports in **timingReports/** directory, **CHIP_postRoute.slk** shows the timing analysis results. All slack values must be positive value in this file. Moreover, for detail path report, see **CHIP_postRoute_all.tarpt**. DRVs report files: ***.cap**, ***.fanout**, and ***.tran**.

**** You can only run postRoute and signoff timing analysis on ee servers ****

7. If the timing slack is negative, or there are DRVs, open ***ECO→Optimize Design…*** in innovus menu
8. Perform post-Route IPO
   - **a.** Design Stage ◆ post-Route
   - **b.** Optimization Type
     - ➢ ◆ Setup
     - ➢ ◆ Design Rule Violations
       - ◆ Max Cap
       - ◆ Max Tran
       - ◆ Max Fanout
   - **c.** Click **OK** button
9. In innovus menu, open ***Timing → Report Timing…***
10. Perform trial route to model the interconnection RC effects
    - **a.** Design Stage ◆ Post-Route
    - **b.** Analysis Type ◆ Hold
    - **c.** Click **OK** button
11. See timing reports in **timingReports/** directory, **CHIP_postRoute_hold.slk** shows the timing analysis results. All slack values must be positive value in this file. Moreover, for detail path report, see **CHIP_postRoute_all_hold.tarpt**. DRVs report files: ***.cap**, ***.fanout**, and ***.tran**.
12. If the timing slack is negative, or there are DRVs, open ***ECO→Optimize Design…*** in innovus menu
13. Perform post-Route IPO
    - **a.** Design Stage ◆ post-Route
    - **b.** Optimization Type
      - ➢ ◆ Hold
      - ➢ ◆ Design Rule Violations
        - ◆ Max Cap
        - ◆ Max Tran
        - ◆ Max Fanout
    - **c.** Click **OK** button
14. Save design as CHIP_postRoute.inn

## 18. Timing Analysis (Signoff)

**- Optional in this Practice**

Signoff timing analysis is similar to postRoute timing analysis. In addition to invoke Quantus QRC Extraction tool to calculate the RC delay as postRoute, **it also consider the SI affect with the signoff RC**.

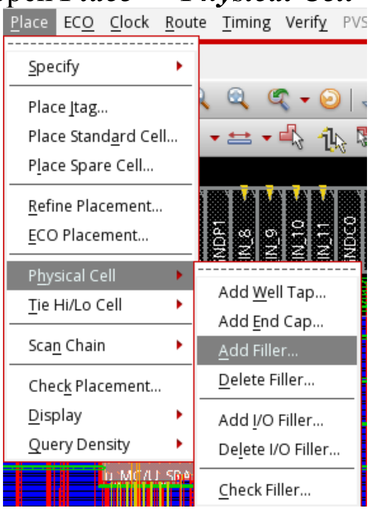
Currently, the delay considering the SI affect with signoff RC can be only used for timing analysis, but cannot be used for timing optimization. If you want to perform timing analysis or timing optimization for postRoute again after signoff timing analysis, you need to reset the SI mode by the command: innovus > **setDelayCalMode -siMode signoff**

1. In innovus menu, open ***Timing → Report Timing…***
2. Perform trial route to model the interconnection RC effects
   - **a.** Design Stage ◆ Sign-Off
   - **b.** Analysis Type ◆ Setup
   - **c.** Click **OK** button
3. See timing reports in **timingReports/** directory, **CHIP_signOff.slk** shows the timing analysis results. All slack values must be positive value in this file. Moreover, for detail path report, see **CHIP_signOff_all.tarpt**. DRVs report files: ***.cap**, ***.fanout**, and ***.tran**.
4. In innovus menu, open ***Timing → Report Timing…***
5. Perform trial route to model the interconnection RC effects
   - **a.** Design Stage ◆ Sign-Off
   - **b.** Analysis Type ◆ Hold
   - **c.** Click **OK** button
6. See timing reports in **timingReports/** directory, **CHIP_signOff_hold.slk** shows the timing analysis results. All slack values must be positive value in this file. Moreover, for detail path report, see **CHIP_signOff_all_hold.tarpt**. DRVs report files: ***.cap**, ***.fanout**, and ***.tran**.

You can perform the above steps to see if there is any difference between the results between the signoff timing analysis and the postRoute timing analysis. **In most experiences and in this practice, signoff timing analysis result is very closed the postRoute timing analysis**. Thus in this practice you can choose to perform this step or not.

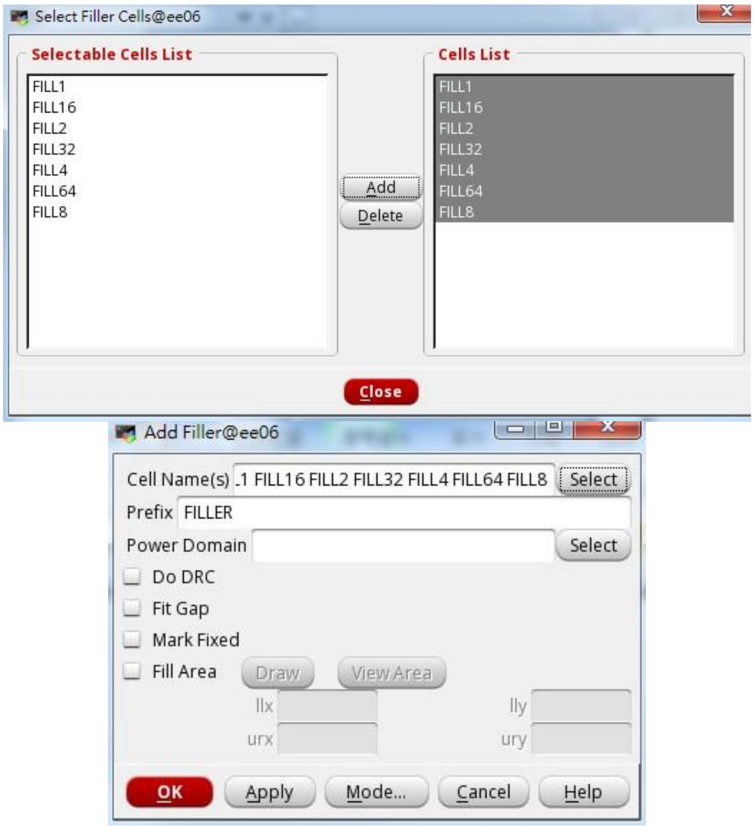
## 19. Add CORE Filler Cells

1. In innovus menu, open ***Place → Physical Cell → Add Filler***




2. Add core filler to improve electric effects of NWELL and PWELL:
   - **a.** Click **Select** button next to Cell Name(s)
   - **b.** Select core filler cells
   - **c.** Click **Add** button
   - **d.** Click **Close** button

**e.** Click **OK** button

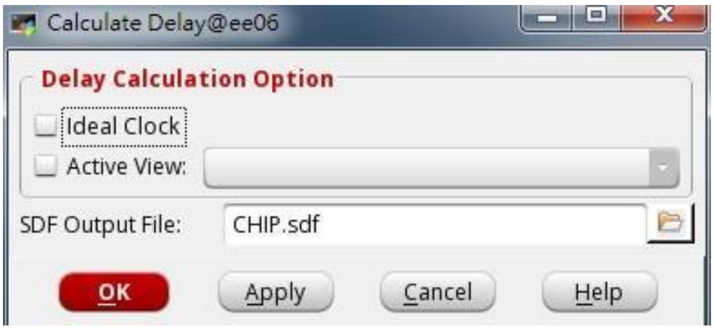



# 20. Stream Out and Write Netlist

1. Save design as CHIP.inn
2. Write CHIP.sdf
   - ***a.*** In innovus menu, open ***Timing → Write SDF***
   - **b.** Delay Calculation Option
     - ➢ ◇ Ideal Clock

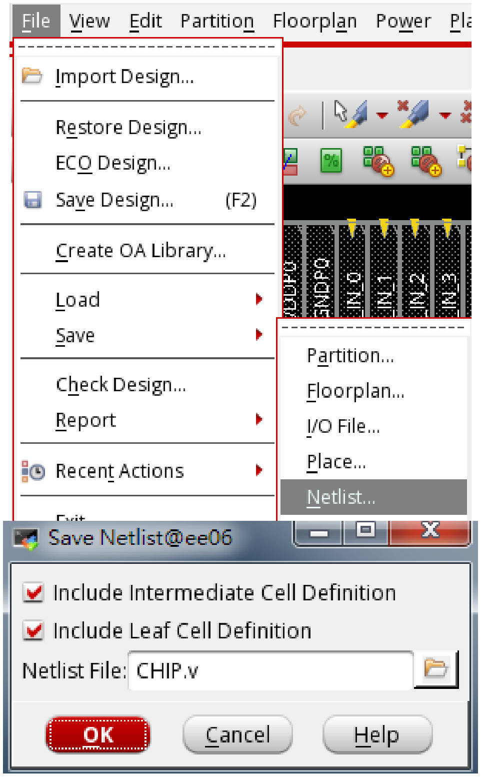
   Ideal Clock should be disabled

   - **c.** Click **OK** button




(or you can use the command “write_sdf CHIP.sdf”)

3. Save design netlist CHIP.v for post-layout simulation:
   - ***a.*** In innovus menu, open ***File→ Save → Netlist…***
     - ➢ ◆ Include Intermediate Cell Definition
     - ➢ ◆ Include Leaf Cell Definition
     - ➢ Netlist File: **CHIP.v**
     - ➢ Click **OK** button




# 21. Post-Layout Gate-Level Simulation

1. Change to directory **06_POST**
2. Perform Post-Layout Gate-level simulation of CHIP.v
   **% ./01_run_vcs_post** The latency should be the same as gate level simulation

## Appendix: IO Pad assignment

CHIP.io:

```
Version: 1
Spacing: 16
Orient: R90
pad: cornerUL                                    NW     CORNERC
Orient: R0
pad: cornerUR                                    NE     CORNERC
Orient: R270
pad: cornerLR                                    SE     CORNERC
Orient: R180
pad: cornerLL                                    SW     CORNERC
pad: opad_QULLR0                                  W
pad: io_vss1                                  W     GNDIOC
.
.
pad: opad_QULLR5                              W
pad: io_vss2                                  W     GNDIOC
pad: io_vdd3                                  N     VCC3IOC
pad: ipad_CODEWORD                            N
.
.
.
pad: io_vdd4                                 N     VCC3IOC
pad: io_vss4                                 N     GNDIOc
pad: opad_QULLR6                              N
```

**You can execute >> source ./cmd/run_apr.cmd in innovus interface to do Step 2. and then you can directly go to Floorplan Step!**